

Business Requirements Document

Automotive Test & Quality Data Integration Dashboard

Field	Details
Document Title	BRD – Automotive Test & Quality Data Integration Dashboard
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Role	Technical Business Analyst (Sample / Practice Project)
Version	1.0
Date	August 2026
Status	Draft – For Demonstration Purposes

1. Introduction

1.1 Purpose of the Document

This Business Requirements Document (BRD) defines the business needs, data sources, system integration points, and functional and non-functional requirements for a centralized Automotive Test & Quality Data Integration Dashboard. It bridges business and engineering stakeholder needs with the technical solution design, serving as the foundation for data architecture, ETL design, and dashboard development.

1.2 Background

Vehicle test and quality data is currently generated across multiple, disconnected systems — engineering test-bench tools, manual inspection sheets, and quality tracking logs maintained by different teams. Engineers and quality analysts manually compile and cross-reference results from these systems to track performance against targets, which is time-consuming, inconsistent in format, and makes trend analysis across vehicle programs difficult.

1.3 Business Objective

- Consolidate engineering test and quality data from multiple systems into a single, structured data source.
- Provide engineers and program managers with a live dashboard to track results against performance and quality targets.
- Reduce manual report compilation time by at least 70%.
- Enable historical trend analysis across vehicle platforms and test iterations.

1.4 Scope

In Scope:

- Data integration from engineering test systems and quality inspection tools (structured export files)
- A centralized data repository for processed test and quality results
- An interactive dashboard (Power BI) showing results against target benchmarks, filterable by vehicle platform, test type, and date
- Automated data refresh on a scheduled basis

Out of Scope:

- Direct real-time integration with test-bench hardware (Phase 1 uses exported files, not live device connections)
- Predictive/ML-based analysis (planned for a future phase)
- Integration with supplier-side test systems

2. Stakeholder Identification

Stakeholder	Role	Interest / Involvement
Test & Validation Engineers	End User / Data Source Owner	Generate test data, need consolidated reporting
Quality Team	Data Source Owner	Provide inspection/quality data, track defect trends
Program Managers	Business Stakeholder	Track performance and quality against targets across programs

Data Engineers	Technical Delivery	Build ETL pipeline and data repository
BI/Dashboard Developer	Technical Delivery	Build and maintain the analytics dashboard
IT Data & Analytics Team	Solution Owner	Own the platform, ensure scalability and governance
Business/Technical Analyst	Requirements Owner	Gather, document, and validate requirements; bridge business and technical teams

3. Current vs. Proposed Data Flow

3.1 Current State (As-Is)

- Engineers and quality analysts export raw results from individual test and inspection systems after each activity.
- Results are manually copied into Excel report templates, one file per test/vehicle/inspection.
- Program-level summaries are manually compiled by cross-referencing multiple Excel files.
- No shared repository — files exist locally on individual team members' machines.

3.2 Proposed State (To-Be)

- Engineers and quality teams export results in a standardized file format to a shared, monitored folder.
- An automated ETL (Extract, Transform, Load) process picks up new files, validates format, and loads structured data into a central database.
- A Power BI dashboard connects to the central database, refreshing on a scheduled basis (e.g., daily).
- Engineers and program managers access the dashboard directly — no manual compilation needed.

3.3 High-Level Data Flow

Source Systems (Test & Quality Tools) → Standardized Export File (CSV) → Shared Data Folder → ETL Process (Validation + Transformation) → Central Database → Power BI Dashboard → End Users (Engineers, Program Managers)

4. Functional Requirements

Req ID	Requirement Description	Priority
FR-01	The system shall accept standardized CSV exports from engineering test systems and quality inspection tools as input sources.	High
FR-02	The system shall validate incoming files for required fields (vehicle ID, test/inspection type, date, measured value, target value) before loading.	High
FR-03	The system shall reject and flag files with missing or malformed data, notifying the responsible engineer or analyst.	Medium
FR-04	The system shall load validated data into a central database, mapped to a defined data schema.	High
FR-05	The dashboard shall display results against defined target benchmarks, with visual pass/fail indicators.	High
FR-06	The dashboard shall allow filtering by vehicle platform, test/inspection type, date range, and responsible team.	High
FR-07	The dashboard shall support historical trend views, showing performance across multiple test/inspection iterations over time.	Medium
FR-08	The system shall refresh dashboard data on a scheduled basis (minimum:	Medium

	once daily).	
FR-09	The system shall maintain an audit log of all data loads, including source file, timestamp, and load status.	Low
FR-10	The dashboard shall support export of filtered views to Excel/PDF for offline reporting.	Low

5. Non-Functional & Technical Requirements

NFR ID	Requirement Description
NFR-01	The ETL process shall complete a standard daily data load within 30 minutes.
NFR-02	The central database shall support at least 5 years of historical data without performance degradation.
NFR-03	The dashboard shall load within 5 seconds for standard filtered views.
NFR-04	The system shall maintain data integrity — no partial/duplicate loads in case of a failed ETL run.
NFR-05	Access to the dashboard shall be role-based (Engineers: view own program data; Program Managers: view all programs).
NFR-06	All data shall be retained in compliance with the organization's data retention and quality audit policies.

6. Assumptions and Constraints

6.1 Assumptions

- Engineers and quality teams will export data in the agreed standardized file format consistently.
- Existing test and quality systems support structured CSV export without additional licensing.
- A shared network location accessible to both source teams and the ETL process is available.

6.2 Constraints

- Phase 1 does not include direct/live integration with test-bench or inspection hardware — file-based export only.
- Solution must use existing IT infrastructure (Power BI, existing database licenses); no new platform procurement in Phase 1.
- Rollout must align with the start of the next vehicle program cycle.

7. Acceptance Criteria (Sample)

For FR-02 (File Validation):

- Given an engineer uploads a CSV export with all required fields populated, when the ETL process runs, then the data should load successfully into the central database with a 'Success' status in the audit log.
- Given a CSV export is missing a required field (e.g., target value), when the ETL process runs, then the file should be rejected, logged with a 'Failed – Missing Field' status, and the responsible engineer or analyst notified by email.

8. Glossary

Term	Definition
BRD	Business Requirements Document
ETL	Extract, Transform, Load — the process of moving and structuring data from source to destination

FR / NFR	Functional Requirement / Non-Functional Requirement
Source Systems	Engineering test and quality inspection tools used to generate raw vehicle data

9. Document Approval

Name	Role	Signature / Approval Status
[Stakeholder Name]	Program Manager	Pending
[Stakeholder Name]	IT Data & Analytics Lead	Pending
Prashanth M	Technical Business Analyst	Prepared

Note: This is a sample/practice Business Requirements Document created to demonstrate Technical BA skills — including data flow design, system integration awareness, and requirements structuring — as part of a professional portfolio. Stakeholder names and organizational details are placeholders.